Testing of transistors

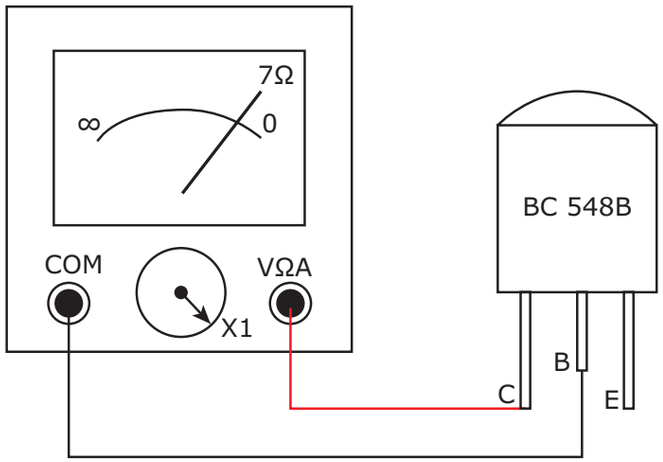


Fig 7.2 Transistor testing

Testing of transistor

Take an NPN transistor.

1. Select X1 in Ohm's range and set 0 Ω adjustment.
   Connect Base with black test probe and Emitter with Red test probe. If the meter reading is approximately 7 Ω, the transistor is said to be in ***good*** condition.

2. Connect Base with black test probe and collector with Red test probe. If the meter reading is approximately 7 Ω, the transistor is said to be in ***good*** condition.

   Now, select X 1K in Ohm's range and set 0 Ω adjustment.

3. Connect Base with Red test probe and collector with Black test probe. If the needle remains unmoved, that is, meter reading is ∞ Ω, the transistor is said to be in ***good*** condition.

4. Connect Base with Red test probe and Emitter with Black test probe. If the needle remains unmoved, that is, meter reading is ∞ Ω, the transistor is said to be in ***good*** condition.

5. Connect Collector with Red test probe and Emitter with Black test probe. If the needle remains unmoved, that is, meter reading is ∞ Ω, the transistor is said to be in ***good*** condition.

6. Connect Collector with Black test probe and Emitter with Red test probe. If the needle remains unmoved, i.e., meter reading is ∞ Ω, the transistor is said to be in ***good*** condition.

   In the above test 3 to 6 , if the meter reading is 0 Ω, the transistor is said to be ***short***.

   In the above test 3 to 6 , if the meter reading shows some ohms such as 100 Ω or 2 kΩ, the transistor is said to be ***leaky***.

   For a PNP transistor, change the polarities while testing.

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